

Did the Beatles Write Less about Love over Time?: An Analysis of 13 Albums from 1963-1970

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Introduction

The Beatles are a band that have heavily influenced music since their first mainstream album in 1963. Because of this, questions surrounding their musical content have been common for decades. Many of these questions are difficult to answer with scientific and statistical backing. This paper aims to give an analytical answer to one question of this kind: were fewer Beatles songs written about love over the course of their career? To address this question, a public dataset obtained from Kaggle was used to analyze thousands of Beatles lyrics.

Data sample

The dataset contains information on 213 different Beatles songs from 17 different albums. The dataset catalogues many different aspects of these songs, but for this analysis only four were used.

- **album**: the name of the album each song was released on
- **year**: the year the album was released
- **word_count**: the total word count per song
- **lyrics**: the full lyrics of the song

The lyrics in this dataset were written in full, i.e. if a chorus is repeated 3 times in the song, the lyrics column in the dataset reflects this. For the sake of usable data, cleaning was required to remove albums which were not being analyzed and to hard code the year of each album, since different songs on the same album often listed different release years.

Results and discussion

A set of target words related to love were chosen and searched for throughout each Beatles album. These words were "love", "loves", "loving", "loved", and "lover".

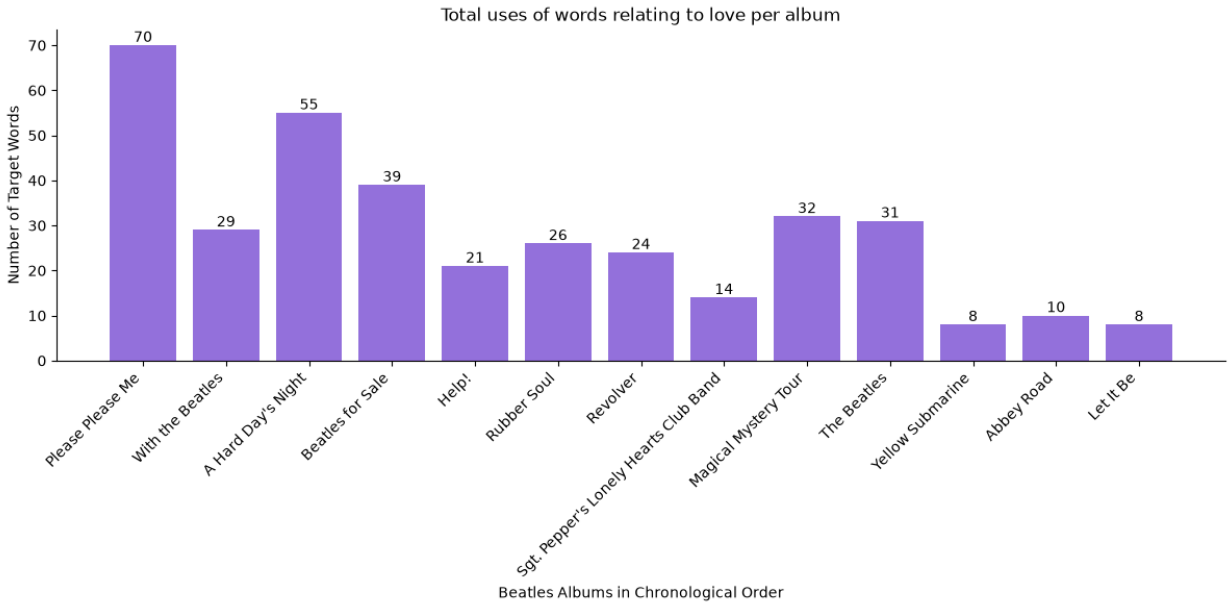


Figure 1: Number of Target Lyrics per Beatles Album

Figure 1 shows the results of a basic compiling of each occurrence of a target word. This gives us a general idea for the mentions of the target words per album, but since each album contains a different number of songs, a more accurate representation would be a percentage.

$$\% = \frac{\#\text{ of target words}}{\text{total word count}} * 100\%$$

We can also calculate the standard deviation, σ, between percent of target words per song on the same album by using the equation for standard deviation,

$$\sigma = \sqrt{\frac{1}{N}\sum_{n_{i=1}}^{N}(x_i - \mu)^2}$$

the individual values, x_i are determined by the percent of target words per song, and the average, μ, is the average percent of target words for the entire album.

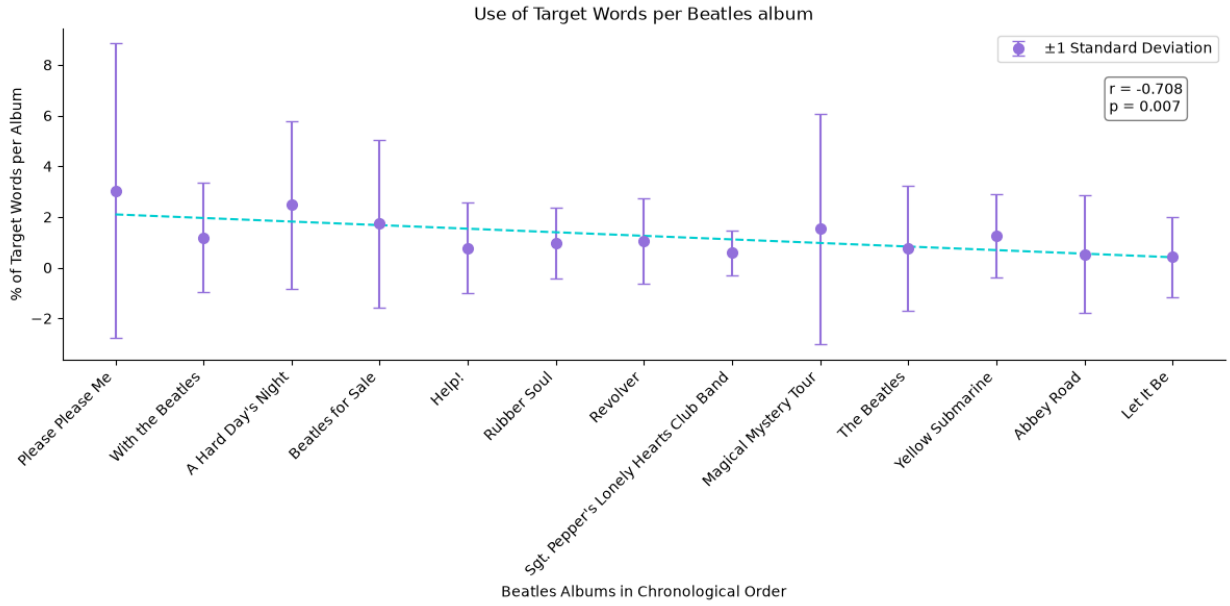


Figure 2: Percent of Target Lyrics per Beatles Album

Figure 2 shows a scatter plot of percent of target words per album, the calculated standard deviations, and the Pearson correlation coefficient, r, and p-value, p.

The standard deviation on certain albums is quite high, suggesting that individual songs on these albums have much higher percentages of target word occurrence than the average song. The very low p-value suggests that it is likely that there is a correlation between the studied variables. The value of r also suggests there is a measurable negative correlation between the two variables.

From this we can conclude that, by these measures, the Beatles did use the word "love" and words related to it less over time. However, the proposed question might take more research to appropriately and fully answer, as one could write about love without using words related to it. Regardless, there is a decrease in obvious mentions of love in Beatles albums over the years.

References

[1] *The Beatles Discography & Lyrics Dataset*. Available at: https://www.kaggle.com/datasets/larissapavan/the-beatles-discography-and-lyrics-dataset.